# GIRMOS grating test project – Progress report

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**Abstract**

Gratings are one of the critical components of GIRMOS which can split a multi-wavelength compact light into single-wavelengths by interference. We tested the optical properties of two transmission diffraction gratings: VPH grating and lithographic grating which were most similar as the actual gratings which will be used in GIRMOS. The light distortions and amount of light passing through the gratings were tested with wavefront error test and efficiency test. For an approximately 10mm diameter beam, the wavefront error PV for VPH grating is $170.7 \pm 11$nm, for lithographic grating is $110.3 \pm 8$nm. The efficiency for VPH grating for wavelength range from 1200 to 1400nm is above 60%. A potential upgrade of the system such as: replace the collimating system with an OAP to allow a bigger aperture; change the lens to one with smaller focal length to ensure all lights go into the detector portal; and add a beam splitter to prevent from unstable light source.

## 1 Introduction

The Gemini Infrared Multi-Object Spectrograph (GIRMOS) is an adaptive optics-fed multi-object integral field spectrograph with a parallel imaging capability (Kchiboucas, 2021). The target astronomical objects for GIRMOS are primordial galaxies with redshift range $1 < z < 10$. We are aiming to study the formation, evolution and characteristics of those targeting galaxies by capturing and analyzing their spectrum to determine their chemical components.

Gratings as one of the most essential part of the spectrograph, which break down the compacted multi-wavelength light into single wavelength lights. It involves light interference and diffract light into targeting direction. Our motivation to do this study is we would like to see how much distortion of the light is given from the grating, and how much light can pass through the grating and diffract to the expected direction. To answer the above two questions, we will perform the wavefront error test and the efficiency test. Our goal is to test out if the wavefront distortion is in an acceptable range, and the efficiency is as high as the given value from the manufacture. Because we want to maximize the amount of light passing through the system, every single photon contains a lot of information, and we do not want to miss any of them. The results from these test will later on help us to adjust the bias and improve the system in advance.

The report is formatted as following: Later this section (Section 1) will introduce all relevant background information related to this experiment including the gratings and tests. Section 2 introduced the model, property, function of each instrument in detail. Section 3 explained the experiment set up and the testing procedures for both wavefront error and efficiency test. Followed by an analysis of the results obtained from the tests in Section 4. And lastly, Section 5 concludes the main results and discussed possible improvements and possible future works.

## 1.1 Diffraction grating

Diffraction grating is a tool to split the light by its wavelengths. It is a regulated structure which involves the principle of multi-slits interference effect, so that the amplitude and the phase of the incident light can be spacially adjust. The diffracted angle can be relate to the wavelength by the following equation:

$$m\lambda = d(\sin(\theta) + \sin(\theta')) \tag{1}$$

Where $m$ is the grating order, $\lambda$ is the wavelength, $d$ is the groove distance of the grating, $\theta$ is the incident angle, and $\theta'$ is the out-coming angle (Palmer, 2020).

The surface of the gratings are normally designed into grooves with periodic shapes. Different grooving shape and grooving distance will causing a different diffraction angle of different wavelength. The two major types diffraction gratings are reflective and transmissive named after whether the light got reflected as with a mirror or transmitted as with a glass.

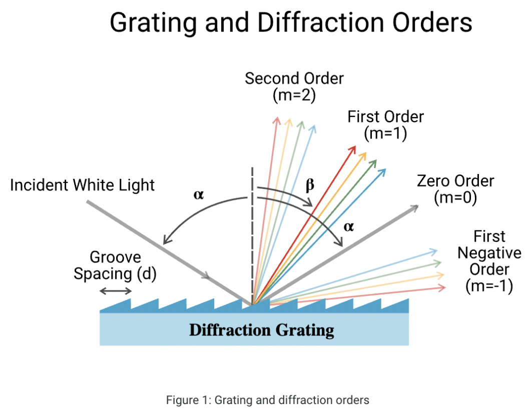


Figure 1: Sample reflecting diffraction grating with triangular grooves. Sourced from meetoptics (Meetoptics, 2025)

Use figure 1 as an example, grating order $m$ are integers representing the propagation-mode of interest. Depending on the out-coming angles, we conclude them into three groups: zero order when $m = 0$, the out-coming angle is the same as the incident angle, and no splitting on wavelength. Positive orders $m > 0$, where the out-coming angles are smaller than the incident angle. Light were successfully split into individual wavelengths. And negative orders where $m < 0$, out-coming angles greater than incident angles. Most of the gratings where designed so that at a specific incident angle, the first order $m = 1$ will have the greatest intensity and highest efficiency.

We will be using the the following two specific gratings: VPH grating and lithographic grating. Consider the time and money budget, we choose the gratings which are most similar to the actual grating that will be used in GIRMOS imager system. Rather than spending time and money to customize one, we can test out if the values matches the one provided from the manufacture, and determine if the supplier is reliable.

### 1.1.1 VPH grating

"Volume Phased Holography (VPH)" gratings have sinusoidal-shaped grooves which were made of photoresist gel and placed in between two glasses. It was constructed by the interference of two sets of waves. The sinusoidal-shaped grooves allow more light pass through and diffract. Less scatters will occur as all the sharp edges are removed, which leads to a decrease of light dispersion, the abrasion is also decreased. The one we are using in our test has 600L/mm groove density, and specifically designed for 1140 to 1450nm. The designed incident angle is 20.4 degree.

The advantage of using VPH grating is that they have a very standard, smooth efficiency curve covers a wide range of wavelength. For the wavelength range mentioned above, the minimum efficiency is 70%, and the maximum efficiency is 90%. Due to the choice of material and the grooving shape, VPH gratings have higher accuracy compare to the ruled gratings. They better resolved the re-imaging and scattering problems as well as a low sensitivity to polarization (Photonics, 2025). Moreover, The glass layers will also help prevent from scratches and damages, and extend the using time. This will raise the mechanical stability as the glasses reduce the change of the grating shape. Which make VPH gratings especially suitable for high resolution astronomical usage.

#### 1.1.2 Lithographic grating

Another grating that we will be testing is the lithographic grating. It involves lithographic and etching techniques, where by e-beam or UV lithography, the photoresist gel were masked and exposed with the expected shape onto a silicon or fused silica substrate. The depth of the notches will control the grating efficiency (Gail, 2008). The groove can be shaped into triangular, rectangular or customized shapes, but as they kept the sharp edges, which means there might be a small possibility to have scatters. We will be using T-940CL, which is a transmission diffraction grating with 940.7L/mm, adaptive to 1526-1610nm with a designed incident angle of 47.5 degree.

Similar as the VPH gratings, lithographic gratings also follow a very smooth efficiency curve. With the specific targeting wavelength range, the minimum efficiency can reach 92%, and the maximum can reach about 96%. With this high efficiency, we can expect less light loss and a better transformation of photon/galaxy information. In terms of the mechanical stability, since there is no protecting layers outside the grating, the usage time might be slightly shorter as it might be hard to prevent from scratches and damages. By the high efficiency and low dispersion, it is also a good candidate for precise astronomy instruments.

### 1.2 Wavefront error test

Wavefront are the field with same phase, which is usually perpendicular to the direction where light is traveling. For example, the wavefront of a parallel beam traveling horizontally would be vertical, the wavefront for a divergent light would be spherical. Wavefront error measures the distortion due to imperfection of the grating and material defects (telescope optics, 2024). The distortion is showed as the displacement of the wavefront compare with the reference wavefront. This distortion will effect the optical response with different wavelength, and make the spectrum result slightly off from the standard plane wave. We use "Peak-to-Valley (PV)" and "Root Mean Square (RMS)" to quantify the change of wavefront. PV measures the difference between the maximum and minimum amplitude of distortion, and RMS calculates the overall absolute difference with the mean displacement value:

$$RMS = \sqrt{\frac{1}{n}\sum_i x_i^2} \tag{2}$$

It is very important to obtain a good out-coming wavefront for the grating because it is directly relate to the image quality, and the distortion is irreparable. All other components in IMGR system rely on the result of the grating, and as it is located on the pupil, all light will pass though the full aperture of the grating.

### 1.3 Efficiency test

Grating efficiency is the percentage of incident light diffracted into specific order (ADE et al., 2021). We calculate it by taking the ratio between the intensity of the incident light and the out-coming light. Since the signal from outer space would be very faint, maintain a high Signal-to-noise ratio is necessary. The higher the applicable efficiency, the stronger the SNR, which means a clear targeting goal that we can observe. As explained in previous, we are expecting very smooth efficiency curves with very high efficiency, so that we can maximize lights which been captured or analyzed by GIRMOS.

# 2 Equipment

This section will introduce all equipment used in the experiment, the main components are: light source, fiber, collimator, diffraction grating, iris, lens and the detector. The model, property and function of each component will be address in details. The collimator, gratings and lens will be unchanged for both tests, others are not shared between the tests. I will introduce them following the order of the path of light.

## 2.1 Light source & fiber

**Wavefront test**

The light source we will be using for the wavefront test is a 638nm red laser beam with adjustable power. This gives us a fixed, stable single-wavelength light. We connect the laser with a single mode fiber with core radius $50\mu m$ which is adaptive to corresponding wavelength and suitable for transformation of single wavelength light.

**Efficiency test**

We will be using a 300W white light source connected with a CS130B-MONO monochromator which breaks down the white light and only gives us a single wavelength light as the out-coming light. It contains 5 color filters and two diffraction gratings inside, provide light with a wavelength in the range of 280nm to 2500nm. With the 2400L/mm grating inside the monochromator, and accuracy of the wavelength is ±0.15nm, the precision is ±0.0375nm. With the 1200L/mm grating, the accuracy is ±0.25nm, the precision is ±0.075nm. We used a $760\mu m$ shutter to reduce the light dispersion. The fiber we used for this case is M24L01 multi-mode fiber with a core radius of $200\mu m$. It can transfer light with 400-2200nm wavelength with minimum light loss.

## 2.2 Collimator

The other end of the fiber is connected to a collimator system. This system involved two concave mirrors.

As figure 2 shows, the divergent light coming out the fiber will hit the smaller mirror first. This mirror has a focal length of 40mm. It bends the light 90 degrees to let it hit the bigger mirror. The bigger mirror has a focal length of 185mm. The resulting light after reflect from the mirror will become parallel. The aperture of the collimated beam has a diameter around 60mm.

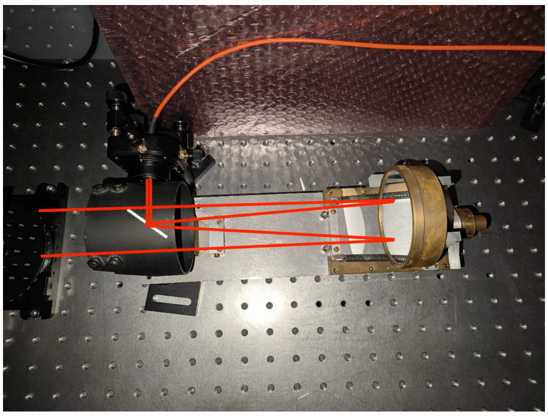

Figure 2: Collimating system and the path of light. The smaller mirror is represented by the white line.

## 2.3 Grating & Iris & Lens

As we previously introduced, we will be using VPH grating and lithographic grating. Our gratings are set on a rotation stage, because we need to rotate the grating into the designed angle to follow the algorithm and allow the best performance. The size of VPH grating is 50.8mm in diameter and 6mm in thickness. The size of lithographic grating is 30x20x4mm. The rotation stage is calibrated so that the norm of the grating holder is perfectly aligned with the zero-degree mark. The focal length of the lens is 200mm and placed parallel with the 0 degree marking.

An iris is placed in between the grating and the lens to only limit a small aperture of light. One reason is that we would like to compare the performance between the two gratings, lithographic grating is very small. Another reason is due to the design of our collimated system, a circle of light in the center was blocked. We would like to avoid this and capture a full light spot, thus we use an iris here.

### 2.4 Wavefront camera

To perform wavefront tests, we will use PHASICS SID4 HR camera. It measures the wavefront of visible to near infrared light with 400 to 1100nm. It performs 416×360 ultra-high phase sampling and 500$\mu m$ PV high dynamic range (Phasics, 2025). The high sensitivity makes it suitable for our testings. The camera has a phase resolution less than 2nm RMS and can process real-time image with frequency at 3fps. When wavefront reaches the camera surface, the camera will pass through a lenslet array, and this array break down the wavefront into a light dot matrix. The dot matrix of plane wave is fixed and uniformed. So by calculating the displacement of the light dots and reduce to gradient, we can obtain the distorted wavefront and further calculate the error.

### 2.5 Photodetector

S148C Integrating sphere photodiode power sensor was used to measure the efficiency. It can detect the current of light with in the range of 1200nm to 2500nm. The power range of this detector is from 1$\mu$W to 1W (Thorlab, 2025). The detector will be placed at the focal point of the lens, so that all light got focused into the detector portal. Light will enter from the entrance portion, then hit the white inner surface, and equally reflect inside the reflecting sphere. As the whole inner surface covered by equal intensity of light, the exist portal will capture the light and perform photoelectric effects. The current will then be recorded and we can further use that to calculate the efficiency.

## 3 Set up and measurements

This section will show the experiment set up for both tests. The alignment process, measuring precedure for both wavefront error test and efficiency test will be explained.

### 3.1 Wavefront error test

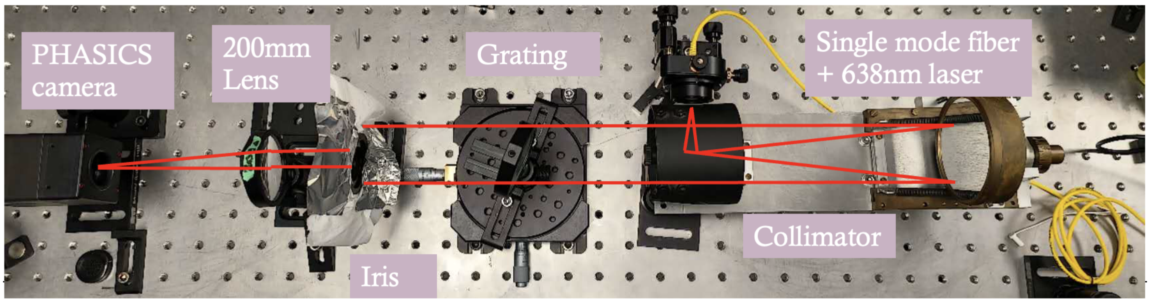


Figure 3: Set up and path of light for wavefront error test.

Figure 3 shows the set up of the wavefront error test, where each component was introduced in previous section. Let's begin with the alignment. The distance between the fiber and the small mirror equals to the focal length of the small concave mirror from the collimator system. The smaller mirror is place on the focal point of the bigger mirror which is fixed in place. We should obtain parallel beam with the size of the bigger concave mirror. To test out is the beam is successfully collimated, we can either measure the diameter of the beam at different distance or use a PSM to see if the light converge to a spot with correct pixels. We will then place an iris after the gratings to limit the size of the beam into a 10mm diameter beam. This size is the maximum size to capture full aperture light passing through two gratings even with high rotation angle. There is no strict limit for the distance for the grating, iris and lens as the beam is parallel. The PHASICS camera is placed a bit closer to the focal point of the lens, because the wavefront will be too hard to measure at the focal point based on the limitation of its size.

We will be using SID4 software which gives us real-time measurement of the wavefront with 3D expression to help visualize. Start the camera on the software, select the mask and circle the bright spot. We will adjust the radius of the mask to make sure all bright spot is included, and avoid any anomaly edge effects. Set an exposure time to make sure there is no dispersion and the light spot is clearly showed on the screen. The PV and RMS is automatically calculated within the system. Remove the grating and keep everything else the same, set this as the reference wavefront. Place the grating onto the rotation stage, and rotate to expected angle. Using acquisition mode to capture 10 continues measurement for both gratings. We then can use python to visualize and analyze the wavefront data.

### 3.2 Efficiency test

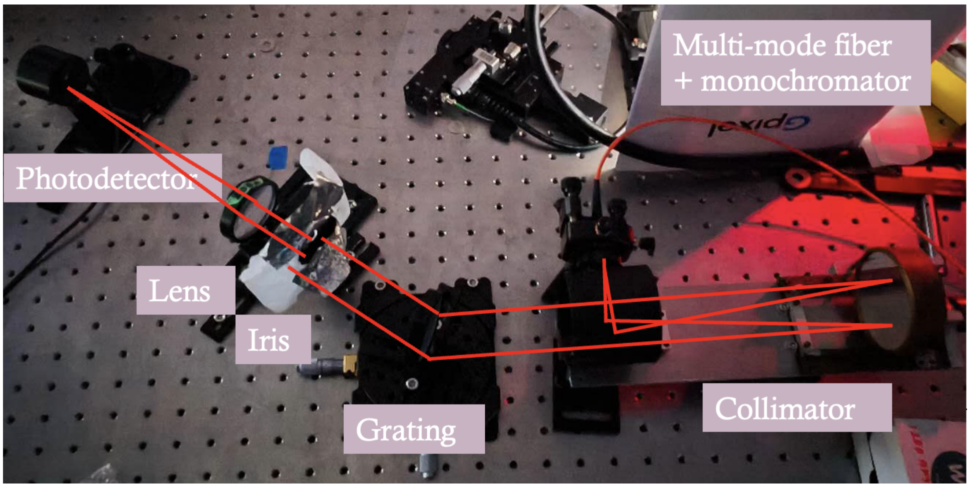


Figure 4: Set up and path of light for efficiency test. Bending angle for the light is 20.4 degress, matching the designed incident angle of VPH grating.

We will use a monochromator with multi-mode fiber and a photodiode detector. The photodetector must placed on the focal point of the lens so that all lights went inside the detector. We will use python to control the wavelength coming out from the monochromator and the detector. The alignment process can be done with visible light then switch to near infrared. Slightly different than the wavefront test, beside the zero order, where all equipment are aligned along a straight line, we will now looking at the first order (m=1). So we will move the iris, lens and photodetector into a corresponding angle which in this case for VPH grating is 20.4 degrees same as the designed incident angle. We will measure the background current – without gratings at zero order, m=0 current – with grating rotated into designed angle and m=1. The efficiency can be simplly calculated by taking the ratio between them. But worth to mention, because the out-coming angle is depend on the wavelength, so for m=1 case, light will not be focused on the same spot. In order to make sure we measured the full range, we might need to move the detector. This will be discussed in later sections.

# 4 Result

With everything successfully set up as introduced in last section, we can start to take measurements and plot out and calculate the results. Two experiment will be discussed separately.

## 4.1 Wavefront error result

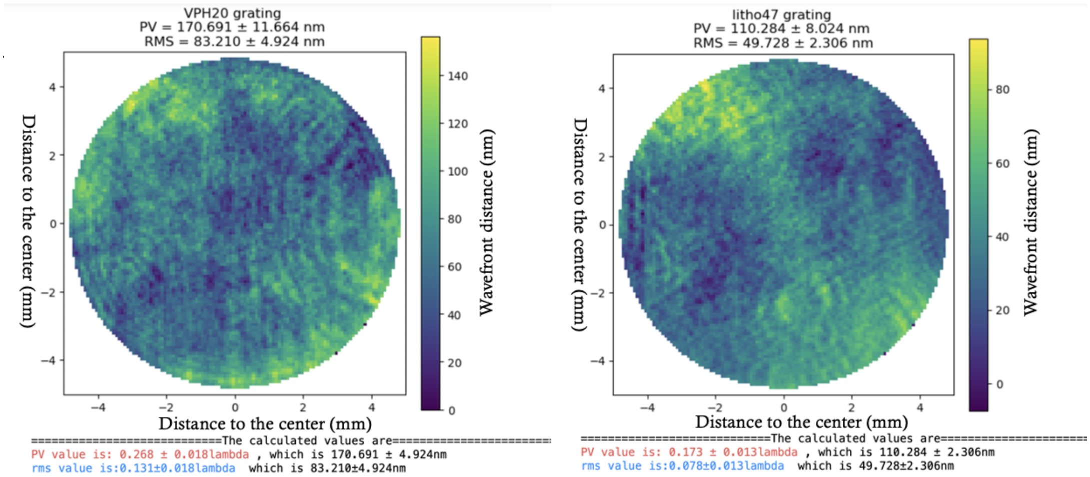


Figure 5: Wavefront error result for VPH grating with 20.4 degrees rotation and lithographic grating with 47.5 degrees rotation. Aperture size is 10mm, limited by the iris.

Figure 5 shows a 2D view of the wavefront for both gratings, where the x and y axes are actual size of the beam in mm, and the color bar is the wavefront displacement in nm. The result shown are the average wavefront error, the VPH grating has a rotation angle of 20.4 degrees, and lithographic grating has a rotation angle of 47.5 degrees. The size of the beam matches the diameter of the iris.

The PV value for VPH grating is 170.7±11nm, came from the difference between maximum and minimum value. RMS is 83.2 ± 4nm calculated using equation 2. PV value for lithographic grating is 110.3±8nm, RMS is 49.7±2nm. We can see from this, that lithographic has a smaller wavefront error comparing with the VPH grating. The RMS value is less than half of PV-value as it calculates the square-rooted mean value. Overall, these values are in an acceptable range, but we are concerning about the size because we only measured a very small diameter of light.

## 4.2 Efficiency result

Figure 6 shows the efficiency of the VPH grating at it's first order. The wavelength range tested is 1200-1400nm. The light red curve at the back is the efficiency data calculated directly taking the ratio of currents. We can see that the result we obtains is very unstable, it is very unlike the smooth curve provided by the manufacture. The darker red curve at the front is a polynomial fit of the original efficiency data. We can see from the fitted line the maximun efficiency is around 1210nm which is 90%, and as the wavelength increase, the efficiency decrease to 60%. The blue points and blue curves are the calibrated points and its fitted line. As mentioned previously, the diffracted light does not focus onto the same spot, which been said if we want to make sure all light goes into the detector, we need to move the detector horizontally to allow it capture the full wavelength range. This will be discussed with more details in the next section as well as possible ways to upgrade the system and avoid this problem.

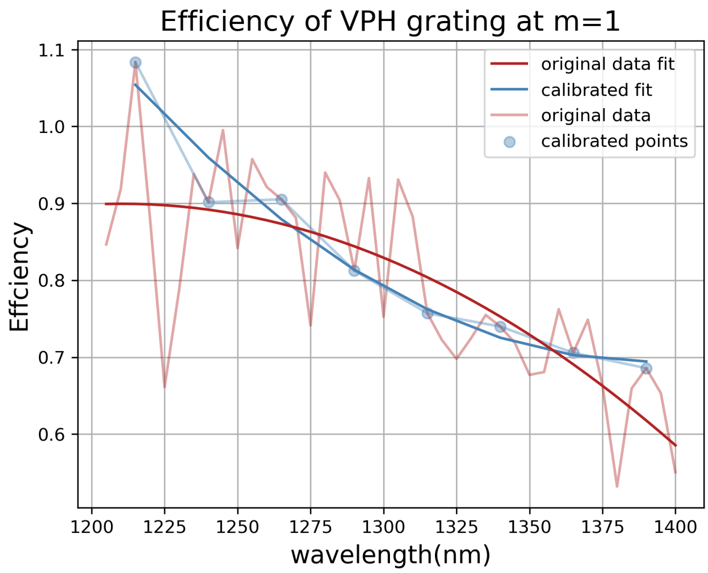


Figure 6: Efficiency curve (light red) and fitted line (dark red) with calibrated point (blue points) and fitted line for calibrated data (dark blue curve).

## 5 Discussion and future work

In conclusion, we obtained the PV-value with a 10mm diameter light spot is $170.7 \pm 11$nn for VPH grating, and $110.3 \pm 8$nm for lithographic grating. Due to limitation of the monochromator, we can only obtain the efficiency for VPH grating. The fitted curve of the efficiency shows that around 1210nm, the efficiency is 90%, and for 1400nm, the efficiency decrease to 60%. Although the wavefront error is in the acceptable range, but the aperture is too small which is not relevant to use. One of the problem we were facing while we do the wavefront test is that, part of the light was blocked by the collimator system. Thus, we cannot capture the full aperture of the light passing through VPH grating. The efficiency result we obtained is very badly expected. Two possible reasons which caused this bad results are: the detector has been moved many times while measuring the current. This might cause some unstableness of the result as we are adding one more variable. Another possible reason might be the unstableness of the light source. As we collected many groups of data, plus the time adjusting the detector, the monochromator might not giving exactly same wavelength all the time. A small shift might result in a big difference. This section will discuss the solutions for above problems, and the future works which can be done for further testings.

For VPH grating, we only took a very small part of light, not fully filling the aperture. To fix this issue, we upgraded the collimator system and replaced by an Off-Axis Parabolic mirror (OAP). OAP perfectly solve the aperture problem as the concave parabolic mirror will bend the divergent light coming in into a parallel beam pointing at the target direction. There will be nothing blocked in-between, so the collimated light can fully cover the VPH grating. Now we can repeat the above measurement for wavefront error and look at how the PV value would change after OAP involves. We are expecting a slight higher PV-value because we increased the diameter about almost 5 times. But the overall wavefront error should still be in the controllable range to give the minimum distortion and minimum affect of the light which will be further analyze.

Recall in Section 4, we've mentioned to move the detector in order to make sure all diffracted light goes into the detector portal.

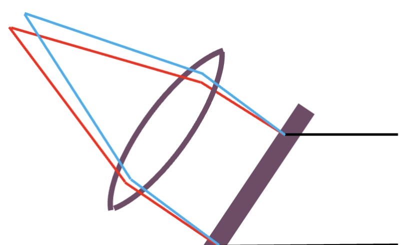

Figure 7 shows a good example where a multi-wavelength light got diffracted by the grating, light with different wavelength will focus onto different locations. We can link this distance with the focal length of the lens and the wavelength using equation 1.

$$dx = d\lambda \frac{fm}{dcos(\theta')} \qquad (3)$$

Where $dx$ is distance between the focused location, $d\lambda$ is the difference of wavelength from the reference wavelength, $f$ is the focal length of the lens, $m$ is the order and $\theta'$ is the out-coming angle from the grating. In our case, we are looking at m=1, so the order is always fixed. $\theta'$ can be calculate from equation 1. So the distance $x$ will be smaller if we use a lens with smaller focal length $f$. And by calculations, if we assume the portal diameter is 5mm, the focal length need to be smaller than 20mm to allow targeting range of wavelength for VPH grating to focus into the detector portal.

In order to solve the unstableness problem of the light source, we can try to place a beam splitter.

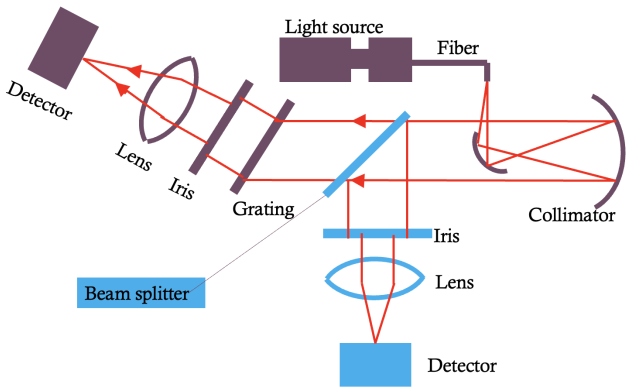


Figure 8: New set up with beam splitter. Increased one path of light.

As figure 8 shows, we can split the collimated beam into two "identical" path with same size of iris, same focal length lens and same detector. We place the grating of one path, so that we can obtain the background current and the current with grating simultaneously, the efficiency can also be calculated right away. This new set up solves the problem of the slight change of the wavelength from the light source, as both detector will receive same change at same time. We can later on replace the collimator system with OAP for the efficiency test and fix the monochromator to test the efficiency for lithographic grating.

If all tests done under room temperature give us acceptable results, we can then do the cold test. We would like to see how the gratings behave under extreme conditions. Since we are sending GIRMOS up to sky, we would like to make sure the gratings have a high adaptability to cold temperatures. Thus, we want to put the system into a vacuum chamber with -193 degree Celsius and repeat the wavefront and efficiency test. A safety test such that pouring liquid Nitrogen onto the gratings will be done in advance before the cold test to make sure the grating is adaptive to extreme temperatures. The cold temperature may damage the grating or glass protecting layers and deform the shape of the grooves. The image quality, diffraction angle, efficiency may all change because of that as mentioned earlier, the wavefront distortions are irreparable. The results obtain from the cold test will be a strong evidence of the feasibility of the optical design, which will give us a big encouragement.

## References


ADE, P. A., J.Griffin, M., and E.Tucker, C. (2021). *Physical Principles of Anstronomical Instruments*. CRC press, 1 edition.

Gail, O. (2008). Lithography: Interference lithography creates 50-nm-period gratings.

Kchiboucas (2021). Gemini infrared multi-object spectrograph (girmos).

Meetoptics (2025). Groove density.

Palmer, C. (2020). *Diffraction Grating Handbook*. MKS Newport, New York, 8 edition.

Phasics (2025). Sid4.

Photonics, W. (2025). Advantages of vph gratings.

telescope optics (2024). Telescope optical aberrations: Types and causes.

Thorlab (2025). Photodiode power sensors (c-series).